

BGFM: BOLTZMANN-CALIBRATED FLOW MATCHING WITH UNIVERSAL NEURAL POTENTIALS FOR 3D MOLECULAR GENERATION

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Paper under double-blind review

ABSTRACT

Fast 3D molecular flow and diffusion models generate diverse de novo molecules, but their probability geometry is learned from empirical molecular datasets rather than from a physical energy landscape. Energy-based molecular generators recover an explicit scalar energy, but sample with expensive Langevin or parallel-tempering chains and can trade diversity for physical quality. We propose **BGFM**, an external-neural-potential calibrated flow generator for de novo 3D molecular generation. BGFM keeps a FlowMol3-style amortized proposal, but trains it with OMol25-derived forces and energies (Levine et al., 2025) through two Boltzmann regularizers: late-time force-score consistency and within-parent density-energy variance. BGFM also learns a calibrated scalar energy head from the same external neural-potential teacher and uses it for a short coordinate-only learned-energy corrector after the flow proposal. The resulting model is designed to combine the speed and diversity of flow matching with the low-strain sampling behavior of energy-based models. Because OMol25 labels enter training, we separate mechanistic OMol25 alignment from external validation: the headline experiments use standard QM9 and GEOM-Drugs (Axelrod and Gómez-Bombarelli, 2022) generation metrics, independent GFN2-xTB (Bannwarth et al., 2019), MMFF (Halgren, 1996), and DFT relaxation oracles, quality–diversity–compute Pareto curves with the Vendi score (Friedman and Dieng, 2023), and shuffled-label negative controls.

1 INTRODUCTION

Three-dimensional molecular generators are increasingly used as the front end of simulation and discovery pipelines. A useful generated structure must not only be chemically valid; it should already be close to a physically plausible region of an energy landscape, otherwise substantial relaxation or molecular dynamics is required before the sample can be used. Current models expose a tradeoff. Flow-matching and diffusion models amortize sampling and produce diverse molecules, but standard training matches empirical data distributions and does not, by construction, enforce Boltzmann relative probabilities under a quantum-level energy model. Energy-based molecular generators learn an explicit scalar potential and can rank, filter, and steer samples, but sampling from a learned energy generally requires many inference-time steps.

BGFM addresses this gap by treating a universal neural potential as an external physical teacher for an amortized molecular flow generator. The model keeps a FlowMol3-style de novo proposal (Dunn and Koes, 2025) over atom types, charges, and coordinates, but adds training-time physical calibration from OMol25. The first calibration term aligns the flow-implied coordinate score with neural-potential forces near the data endpoint. The second estimates a coordinate log-density with a FFJORD integral and enforces that, within a cloud of perturbations of the same parent molecule,

$$\log p_{\theta}^{\text{pos}}(r \mid c) + \beta E_{\text{NP}}(r, c) \quad (1)$$

varies as little as possible. This is a local conditional Boltzmann condition: lower-energy perturbations of the same molecular identity should receive higher probability. BGFM further trains a small scalar energy head to approximate the external neural-potential energy and force, then uses

this head for a short learned-energy Langevin corrector that refines coordinates while holding the sampled molecular identity fixed.

The central comparison is with EBMol-style energy-based generation (Griesbacher et al., 2026). EBMol learns a persistent scalar energy and samples with Mirror-Langevin dynamics and parallel tempering. BGFM takes the complementary route: keep the fast flow proposal, calibrate it with external physical labels, and use a short learned-energy corrector rather than a long MCMC chain. Therefore, our primary evaluation is not just validity or an OMol25 self-correlation. We evaluate whether BGFM improves the physical-quality-diversity-compute frontier under independent relaxation oracles.

Contributions.

1. **External-neural-potential calibrated amortized 3D molecular flow generation.** We propose a single de novo generator that retains a FlowMol3-style amortized proposal and trains it against OMol25 forces and energies as an external physical teacher.
2. **Conditional-coordinate Boltzmann regularization.** We combine late-time force-score consistency with within-parent density-energy variance to align the model’s conditional coordinate density with OMol25-induced Boltzmann relative probabilities, while avoiding the ill-defined claim of a global Boltzmann distribution over arbitrary molecular identities.
3. **Calibrated energy corrector that improves independent physical quality.** BGFM learns a composition-conditioned strain scorer from OMol25 energies and gradients and applies it as a short coordinate-only Langevin corrector. We measure its effect on the quality-diversity-compute frontier against EBMol-style energy-based sampling using independent GFN2-xTB / MMFF / DFT oracles.

2 RELATED WORK

3D molecular flow and diffusion. EDM (Hoogeboom et al., 2022), GeoLDM (Xu et al., 2023), MiDi (Vignac et al., 2023), GeoBFN (Song et al., 2024), SemlaFlow (Irwin et al., 2024), FlowMol (Dunn and Koes, 2024), and FlowMol3 (Dunn and Koes, 2025) learn transport or denoising maps for molecular atom types and coordinates. GeoDiff (Xu et al., 2022) targets conditional conformer generation. The underlying generative families — denoising diffusion (Ho et al., 2020), score SDE (Song et al., 2021), neural ODEs and FFJORD-style flow density (Chen et al., 2018; Grathwohl et al., 2019), and flow matching (Lipman et al., 2023) — supply the velocity / score primitives we reuse. FlowMol3 is the closest backbone baseline: an open-source multi-modal flow-matching model for de novo all-atom small-molecule generation that improves FlowMol via self-conditioning, fake atoms, and train-time geometry distortion without changing the underlying GNN or flow formulation. BGFM is not a replacement backbone; it is an external-neural-potential calibration layer on top of this family of amortized generators.

Energy-based molecular generation. EBMol (Griesbacher et al., 2026) learns a time-unconditional atom-additive scalar potential and samples it with Mirror-Langevin dynamics plus parallel tempering. It is therefore the direct competitor for physically motivated 3D generation. BGFM differs in two ways. First, BGFM uses an external universal neural potential as the physical teacher, whereas EBMol learns an energy from geometry data without direct energy supervision. Second, BGFM retains an amortized flow proposal and uses a short corrector, whereas EBMol relies on inference-time energy sampling.

Flexible conformer ensemble generation. FlexiFlow (Tedoldi et al., 2025) extends flow matching to jointly sample molecular graphs and multiple conformations and reports strong QM9/GEOM-Drugs performance. FlexiFlow is a required standard-generation baseline, especially for conformer-ensemble metrics. Its goal is complementary to BGFM: FlexiFlow focuses on molecular/conformer ensemble coverage, while BGFM focuses on external-neural-potential calibrated low-strain de novo generation.

Boltzmann generators and energy-based flow matching. Boltzmann Generators (Noé et al., 2019), equivariant flow matching for Boltzmann sampling (Klein and Noé, 2023), and iEFM (Woo

and Ahn, 2024) study sampling from known or learned unnormalized densities, often for fixed systems or continuous toy/physical potentials. iEFM is especially relevant theoretically because it trains CNFs from energy function evaluations, but its evaluation setting is fixed continuous Boltzmann sampling rather than mixed discrete-continuous de novo molecule generation.

Universal neural potentials. OMol25 (Levine et al., 2025) provides large-scale DFT data and neural-potential models over a broad chemistry space. BGFM uses such a neural potential not as a post-hoc relaxer but as a training-time teacher for forces, energies, and energy-head calibration.

3 METHOD

Let a molecular configuration be $x = (c, r)$, where c denotes molecular identity variables (atom count, atom types, and charges) and $r \in \mathbb{R}^{3N}$ denotes coordinates. The external neural potential returns an energy $E_{\text{NP}}(r, c)$ and forces $F_{\text{NP}}(r, c) = -\nabla_r E_{\text{NP}}(r, c)$. BGFM does not define a global canonical ensemble over arbitrary molecular identities. Instead, for a fixed identity c , it targets the conditional coordinate Boltzmann law

$$p_B(r \mid c, T) = \frac{1}{Z_c(T)} \exp[-\beta E_{\text{NP}}(r, c)], \quad \beta = 1/(kT). \quad (2)$$

This target is well-defined because c is fixed; cross-composition probabilities are left to the empirical proposal distribution and are not interpreted as thermodynamic weights.

3.1 FLOW PROPOSAL

BGFM starts from a FlowMol3-style flow proposal. Continuous coordinates use a Gaussian prior and a time-dependent velocity field $v_\theta(x_t, t)$; atom types and charges use the categorical flow/CTMC component of the backbone. The base training loss is the standard flow-matching objective

$$\mathcal{L}_{\text{FM}} = \mathbb{E} [\|v_\theta(x_t, t) - u_t\|^2 + \mathcal{L}_{\text{disc}}]. \quad (3)$$

The proposal is responsible for chemical diversity and fast amortized sampling. The physical modules below calibrate this proposal rather than replace it.

3.2 LATE-TIME FORCE-SCORE CONSISTENCY

For the conditional Boltzmann law in Eq. equation 2,

$$\nabla_r \log p_B(r \mid c, T) = \beta F_{\text{NP}}(r, c). \quad (4)$$

Under the linear Gaussian interpolant, the coordinate score at time t can be estimated from the flow velocity by

$$s_\theta(x_t, t) = \frac{t v_\theta(x_t, t) - x_t}{(1-t)\sigma_0^2}. \quad (5)$$

BGFM evaluates this score at late but finite times $t \in \mathcal{T}$ and aligns it with the endpoint neural-potential force:

$$\mathcal{L}_{\text{force}} = \mathbb{E}_{t \in \mathcal{T}} [1 - \cos(s_\theta(x_t, t), \beta F_{\text{NP}}(r_1, c))]. \quad (6)$$

This is a late-time regularizer, not an exact identity at arbitrary intermediate x_t . It avoids online neural-potential calls during every training step by using precomputed endpoint forces.

3.3 DENSITY-ENERGY VARIANCE

The most direct probability calibration is that $\log p_B(r \mid c, T) + \beta E_{\text{NP}}(r, c)$ equals a constant depending only on c and T . BGFM estimates the coordinate log-density $\log p_\theta^{\text{pos}}(r \mid c)$ with a FFJORD-style divergence integral through the coordinate velocity field. For each parent molecule m , we precompute a cloud of perturbations $\{r_m^{(k)}\}_{k=1}^K$ and neural-potential energies. The loss is

$$\mathcal{L}_{\text{dens}} = \mathbb{E}_m \text{Var}_k [\log p_\theta^{\text{pos}}(r_m^{(k)} \mid c_m) + \beta E_{\text{NP}}(r_m^{(k)}, c_m)]. \quad (7)$$

Because the variance is taken within the same parent molecule, the unknown offset $-\log Z_{c_m}(T)$ cancels. We optionally add a composition-conditioned scalar offset $b_\phi(c)$ as an anchor, but do not interpret it as a supervised thermodynamic partition-function estimator.

3.4 CALIBRATED COMPOSITION-CONDITIONED STRAIN SCORER

To obtain an explicit inference-time energy without running OMol25 at sampling time, BGFM trains a scalar head. Absolute OMol25 energies are dominated by composition and atom count, so a direct regression head would mostly learn molecular size rather than coordinate-dependent strain. We therefore decompose the head into a *composition baseline* $b_\phi(c)$ and a *coordinate-dependent strain term* $\Delta E_\psi(r, c)$:

$$\hat{E}_\psi(r, c) = b_\phi(c) + \Delta E_\psi(r, c), \quad (8)$$

and train against OMol25 with the residual loss

$$\mathcal{L}_{\text{head}} = |\Delta E_\psi(r, c) - (E_{\text{NP}}(r, c) - b_\phi(c))| + \lambda_F \|\nabla_r \Delta E_\psi(r, c) - F_{\text{NP}}(r, c)\|_2^2, \quad (9)$$

where b_ϕ is a small invariant network on the molecular composition vector (atom-type histogram + total charge) and ΔE_ψ is the per-graph readout of the head from the flow-proposal scalar features. We use the head as a composition-conditioned strain scorer for ranking and refinement; it is not a free energy and is not a thermodynamic absolute formation energy.

3.5 SHORT LEARNED-ENERGY CORRECTOR

After the flow proposal samples (c, r^0) , BGFM optionally refines only coordinates using the calibrated head:

$$r^{j+1} = \Pi[r^j + \eta_j(-\nabla_r \Delta E_\psi(r^j, c)) + \sqrt{2\eta_j/\beta} \epsilon_j], \quad j = 0, \dots, J-1. \quad (10)$$

Here $J \in \{0, 20, 50, 100\}$ in our compute sweep, Π removes center-of-mass drift, and the discrete identity c — atom counts, types, and charges — is held fixed throughout refinement. Per sampled molecule we report flow NFEs, energy-head NFEs, external oracle calls (zero by default), wall-clock time, and outcome. Samples whose post-corrector geometry produces invalid OpenBabel/RDKit bond inference, steric clash above a threshold, or downstream relaxation failure are counted as *failures* and remain in the population averages of the all-samples protocol; we never silently drop them. Unless a Metropolis correction is explicitly enabled and reported, this corrector is an unadjusted short refinement, not an exact MCMC sampler.

3.6 TRAINING OBJECTIVE

The full training loss is

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_1 \mathcal{L}_{\text{force}} + \lambda_2 \mathcal{L}_{\text{dens}} + \lambda_3 \mathcal{L}_{\text{anchor}} + \lambda_4 \mathcal{L}_{\text{head}}. \quad (11)$$

All auxiliary weights use warmup/ramp schedules. OMol25 is not called in the main training loop: endpoint forces are stored in the dataset and perturbation energies are precomputed offline.

Optional joint-density extension. BGFM can also add discrete log-probability terms for atom counts, atom types, and charges to form $\log p_\theta(x) = \log p_\theta^{\text{pos}}(r | c) + \log p_\theta(c)$. This extension is useful for diagnostics and future full-joint density calibration. Unless explicitly enabled and verified in an ablation, the main objective in this paper should be interpreted as conditional coordinate Boltzmann calibration.

4 EXPERIMENTS

Experimental questions. We design experiments to answer six questions. First, does BGFM preserve standard de novo 3D generation quality? Second, does it improve physical quality under independent evaluators rather than only under the OMol25 labels used for training? Third, does it achieve a better quality-diversity-compute frontier than EBMol-style energy sampling? Fourth, does the proposed objective actually produce local conditional Boltzmann alignment? Fifth, do the gains require correct physical labels rather than arbitrary regularization? Sixth, does the calibrated energy head rank and filter molecular strain better than existing energy scorers?

Datasets and baselines. We evaluate on QM9, GEOM-Drugs, and held-out OMol25 chemistry slices. We report each baseline along with its post-processing variant, so the calibrated-corrector contribution of BGFM is not confounded with generic post-hoc refinement. Specifically, our standard-generation baselines include:

- the official FlowMol3 (Dunn and Koes, 2025) checkpoint with reported numbers and a re-run on our hardware;
- a *same-backbone FM baseline*: an identical FlowMol3-style architecture trained on the same OMol25 split, without any BGFM regularization or corrector;
- FlowMol3/FM *plus the same corrector budget* as BGFM, where the corrector uses an external xTB or OMol25 oracle instead of $\hat{E}_\psi$;
- FlowMol3/FM *plus an xTB / OMol25 re-ranking head* that filters and re-orders the same sample pool;
- EDM, GeoLDM, GeoBFN, SemlaFlow, FlexiFlow, and EBMol with reported numbers, run under matched NFE budgets where possible.

For BGFM we report the full method and its module ablations: flow-only, force-only, density-only, head-only, BGFM w/o corrector, and full BGFM. These splits let us answer whether BGFM’s gains come from the calibrated corrector or from post-processing accessible to any amortized baseline.

4.1 STANDARD GENERATION QUALITY

We generate 10k samples per model under matched or reported NFE budgets and report atom stability, molecule stability, validity, uniqueness, novelty, Vendi diversity, NFE, and wall-clock time.

Table 1: Standard de novo 3D generation quality. Replace all placeholders before submission.

Model	NFE	Atom stab.	Mol. stab.	Valid	Unique	Novel	Vendi
FlowMol3 (official)	reported	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
Same-backbone FM	[RUN: NFE]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
FlexiFlow	reported	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
EBMol	matched	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
BGFM w/o corrector	matched	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
BGFM full	matched	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]

4.2 INDEPENDENT PHYSICAL QUALITY

Because BGFM uses OMol25 labels during training, OMol25 correlation is not the headline metric. Our main physical-quality table uses independent GFN2-xTB relaxation, MMFF relaxation on organic subsets, and a DFT validation subset. We report both valid-connected-only and all-samples protocols.

Table 2: Independent physical quality under evaluators not used as BGFM’s density-energy target.

Model	Valid conn.	xTB med. ΔE	xTB mean ΔE	p90 ΔE	Relax steps	Failure
EBMol-low	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
EBMol-high	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
FlowMol3 (official)	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
Same-backbone FM	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
FM + corrector (same J)	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
FM + xTB rerank	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
BGFM w/o head	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
BGFM w/o corrector	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
BGFM full	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]

4.3 QUALITY-DIVERSITY-COMPUTE PARETO

We sweep flow NFEs and corrector steps $J \in \{0, 20, 50, 100\}$ and compare to EBMol’s compute-scaled samplers. The key metric is the number of valid, unique, low-strain samples per GPU-hour:

$$\text{VULS}_\tau = \frac{\#\{\text{valid, unique samples with } \Delta E_{\text{xTB}} < \tau\}}{\text{GPU-hour}}. \quad (12)$$

This evaluation targets the main BGFM hypothesis: a fast amortized flow proposal plus a short calibrated corrector should dominate long energy-sampling chains on physical quality, diversity, and cost.

4.4 MECHANISTIC BOLTZMANN ALIGNMENT

For held-out parent molecules and perturbation families, we measure the within-parent correlation and calibration slope between $\log p_\theta^{\text{pos}}(r \mid c)$ and $-\beta E(r, c)$. We report this with OMol25 as a mechanism diagnostic, and repeat it with xTB and a DFT subset to test transfer.

4.5 ABLATIONS AND NEGATIVE CONTROLS

A valid physical improvement should disappear or weaken under corrupted labels. We therefore train or evaluate controls with shuffled forces, shuffled energies, wrong parent grouping, wrong temperature, force norm only, and random scalar regularization.

Table 3: Negative controls separating correct physical supervision from arbitrary regularization.

Model/control	OMol25 align.	xTB med. ΔE	DFT subset	Valid	Vendi
FM baseline	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
BGFM true labels	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
Shuffled force	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
Shuffled energy	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
Wrong parent grouping	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
Wrong kT	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]

4.6 CROSS-MODEL RANKING AND FILTERING

We collect samples from EDM/FlowMol3/EBMol/BGFM and deliberately strained perturbations. We compare EBMol energy, OMol25 energy, BGFM’s calibrated energy head, and the BGFM combined score as predictors of independent xTB/DFT relaxation energy and high-strain failure. Metrics are Pearson, Spearman, AUROC for high-strain detection, and top-decile filtering quality.

5 LIMITATIONS

BGFM inherits OMol25’s coverage and accuracy limitations. The main density loss is conditional-coordinate unless the optional discrete joint-density extension is explicitly enabled. The short corrector is a learned-energy refinement, not an exact MCMC sampler. The energy head is a calibrated scorer distilled from neural-potential energies and forces, not a free-energy estimator. These limitations are evaluated with independent oracles and shuffled-label controls rather than hidden.

6 CONCLUSION

BGFM is designed to bridge two strengths: amortized de novo flow sampling and physically meaningful energy landscapes. By calibrating a fast molecular flow proposal with an external universal neural potential and using a short learned-energy corrector, BGFM targets low-strain, diverse, computationally efficient 3D molecule generation. The decisive empirical question is not whether BGFM improves an OMol25 self-metric, but whether it improves independent physical quality and the quality-diversity-compute frontier against strong flow and energy-based baselines.

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